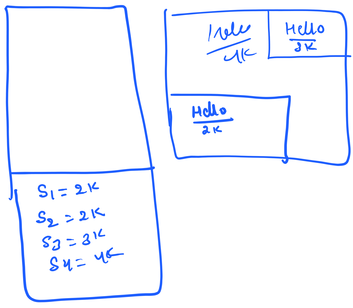
① String → class → NP → heap

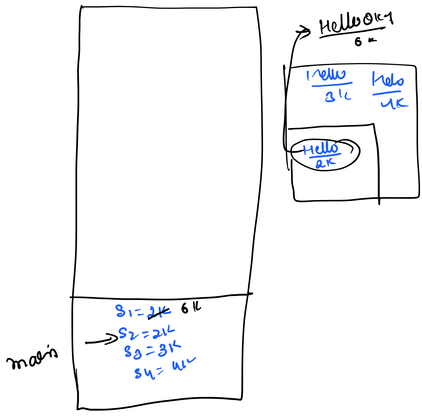
```java
public static void main(String[] args) {
    // TODO Auto-generated method stub
    String s1 = "Hello";
    String s2 = "Hello";
    String s3 = new String("Hello");
    String s4 = new String("Hello");

}
```




```java
public static void main(String[] args) {
    // TODO Auto-generated method stub
    String s1 = "Hello";
    String s2 = "Hello";
    String s3 = new String("Hello");
    String s4 = new String("Hello");
    s1 = s1 + "Okay";
    System.out.println(s1);

}
```



Kunal
Komal

Substring

Hello
0 1 2 3 4

i j
0,1 ← H
0,2 ← He
0,3 ← Hel
0,4 ← Hell
0,5 ← Hello

1,2 ← e
1,3 ← el
1,4 ← ell
1,5 ← ello

2,3 ← l
2,4 ll
2,5 llo

3,4 l
3,5 lo

4,5 o

```
for(i=0  i<s.length() i++){
  for(j=i+1  j<=s.length() j++){
    System.out.println(s.substring(i,j));
  }
}
```

---

i j
[0,1] H
[1,2] e
[2,3] l
[3,4] l
[4,5] o

[0,2] He
[1,3] el
[2,4] ll
[3,5] lo

Hello
0 1 2 3 4

0,3 Hel
1,4 ell
2,5 llo

0,4 Hell
1,5 ello

0,5 Hello

```
for(len=1  len<=s.length() len++){
  for(i=0  j<s.length() i++){
    int j = i+len
    System.out.println(s.substring(i,j));
  }
}
```